PVsyst V8.1.7

PVsyst - Simulation report

Grid-Connected System

Project: Belem

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 6.60 kWp

Belem - Brazil

Author

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Project summary

Geographical Site
Belem
Brazil

Situation

Latitude	-1.4506 °(S)
Longitude	-48.4682 °(W)
Altitude	10 m
Time zone	UTC-3

Project settings
Albedo 0.20

Weather data
Belem
NASA POWER satellite data 2001-2020 - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Horizontal plane

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	11 units	Nb. of units	1 unit
Pnom total	6.60 kWp	Total power	6 kWac
		Pnom ratio	1.10

Results summary

Simulation step	Hourly				
Produced Energy	10671 kWh/year	Specific production	1617 kWh/kWp/year	Perf. Ratio PR	85.22 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Simulation step
Hourly

Orientation #1
Horizontal plane

Models used

Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	JKM-600N-72HL4-V
(Original PVsyst database)	
Unit Nom. Power	600 Wp

Inverter

Manufacturer	Generic
Model	MIN 6000TL-X
(Original PVsyst database)	
Unit Nom. Power	6.00 kWac

Array #1 - PV Array

Number of PV modules	6 units	Number of inverters	1 * MPPT 50% 0.5 unit
Nominal (STC)	3600 Wp	Total power	3.0 kWac
Modules	1 unit x 6 In series		
At operating cond. (50°C)		Operating voltage	80-550 V
Pmpp	3339 Wp	Pnom ratio (DC:AC)	1.20
U mpp	245 V		
I mpp	14 A		

Array #2 - Sub-array #2

Number of PV modules	5 units	Number of inverters	1 * MPPT 50% 0.5 unit
Nominal (STC)	3000 Wp	Total power	3.0 kWac
Modules	1 unit x 5 In series		
At operating cond. (50°C)		Operating voltage	80-550 V
Pmpp	2783 Wp	Pnom ratio (DC:AC)	1.00
U mpp	204 V		
I mpp	14 A		

Total PV power

Nominal (STC)	7 kWp
Total	11 modules
Module area	28.4 m²
Cell area	26.2 m²

Total inverter power

Total power	6 kWac
Number of inverters	1 unit
Pnom ratio	1.10
No power sharing	

Array losses

Thermal Loss factor
Module temperature according to irradiance

Uc (const)	20.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

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DC wiring losses

Global wiring resistance	10.0 mΩ
Loss Fraction	1.50 % at STC

Array #1 - PV Array

Global array res.	293 mΩ
Loss Fraction	1.50 % at STC

Array #2 - Sub-array #2

Global array res.	245 mΩ
Loss Fraction	1.50 % at STC

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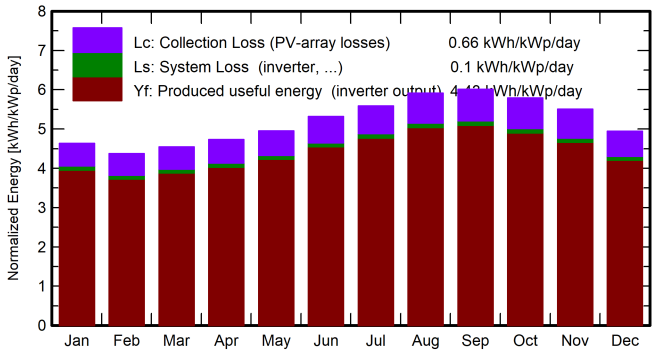
Main results

Simulation step
Hourly

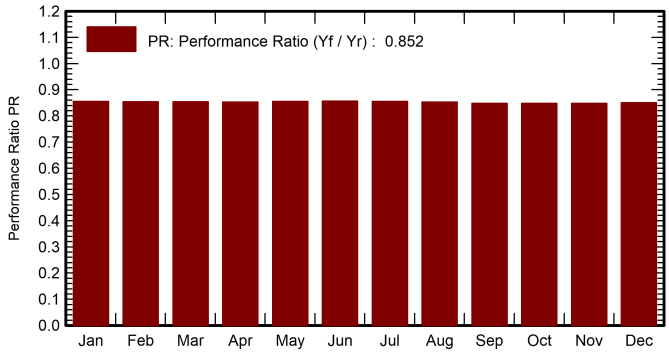
System Production

Produced Energy	10671 kWh/year	Specific production	1617 kWh/kWp/year
		Perf. Ratio PR	85.22 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	143.7	71.70	26.24	143.6	139.9	831	811	0.855
February	122.5	64.89	25.87	122.4	119.4	708	690	0.854
March	141.1	72.74	25.94	141.0	137.5	815	795	0.855
April	142.1	67.83	26.03	142.0	138.6	819	799	0.852
May	153.6	65.17	26.30	153.6	149.6	887	866	0.855
June	159.6	55.54	26.36	159.5	155.5	922	901	0.856
July	173.3	56.41	26.40	173.2	169.1	999	977	0.855
August	183.5	60.51	27.31	183.4	179.4	1055	1032	0.852
September	180.5	64.38	28.47	180.5	176.9	1032	1009	0.847
October	179.5	71.86	28.79	179.5	175.6	1027	1004	0.847
November	165.2	69.50	28.76	165.2	161.3	945	924	0.848
December	153.4	70.28	27.60	153.3	149.3	881	861	0.851
Year	1898.2	790.81	27.01	1897.3	1852.1	10920	10671	0.852

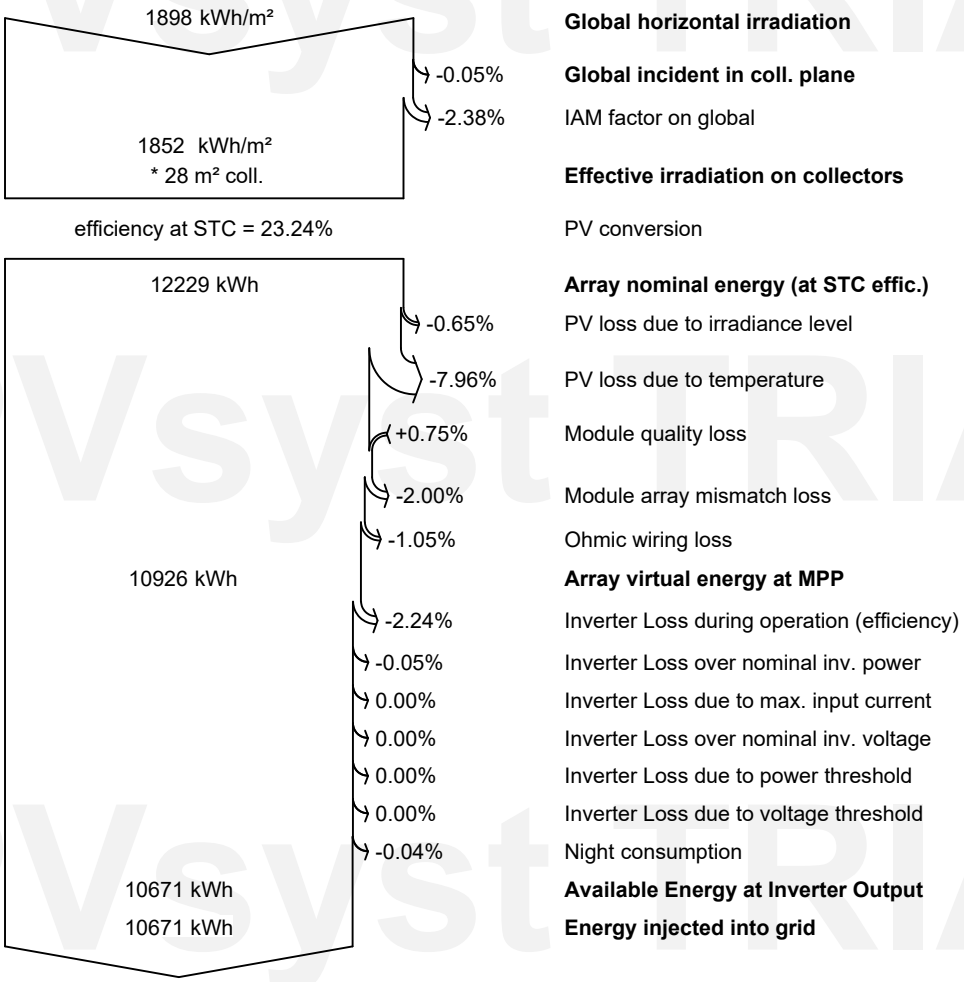
Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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Loss diagram

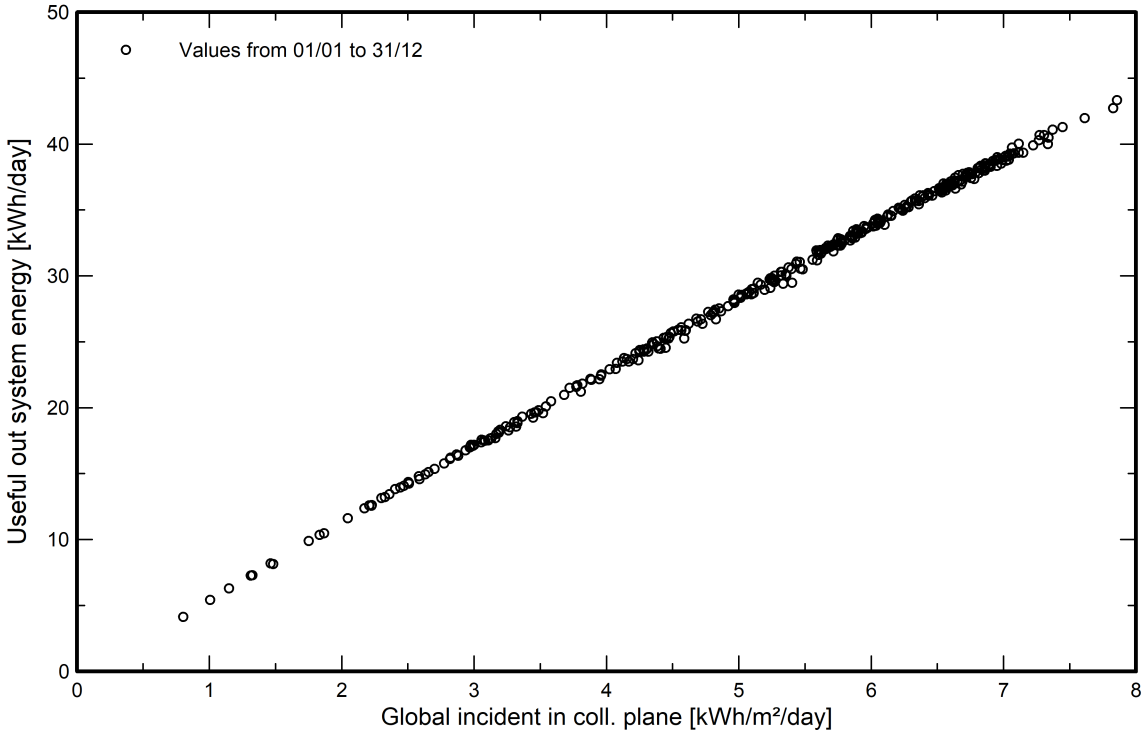


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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

